

Vector Semantics

- Osgood's 1957
 - ✓ use a point in three-dimensional space to represent the connotation of a word
- Joos (1950), Harris (1954), and Firth (1957)
 - ✓ define the meaning of a word by its distribution in language use
 - ✓ two words that occur in very similar distributions (whose neighboring words are similar) have similar meanings.

Vector Semantics

ongchoi

Ongchoi is delicious sauteed with garlic.
Ongchoi is superb over rice.
...ongchoi leaves with salty sauces...

...spinach sauteed with garlic over rice...
...chard stems and leaves are delicious...
...collard greens and other salty leafy greens

Vector Semantics

ongchoi

*ongchoi is a leafy green
similar to other leafy greens.*



Vector Semantics

- **Intuition**

- ✓ represent a word as a point in a multidimensional semantic space that is derived (in ways we'll see) from the distributions of word neighbors.
- Vectors for representing words are called **embeddings**
 - ✓ mapping from one space or structure to another

Vector Semantics



Vector Semantics

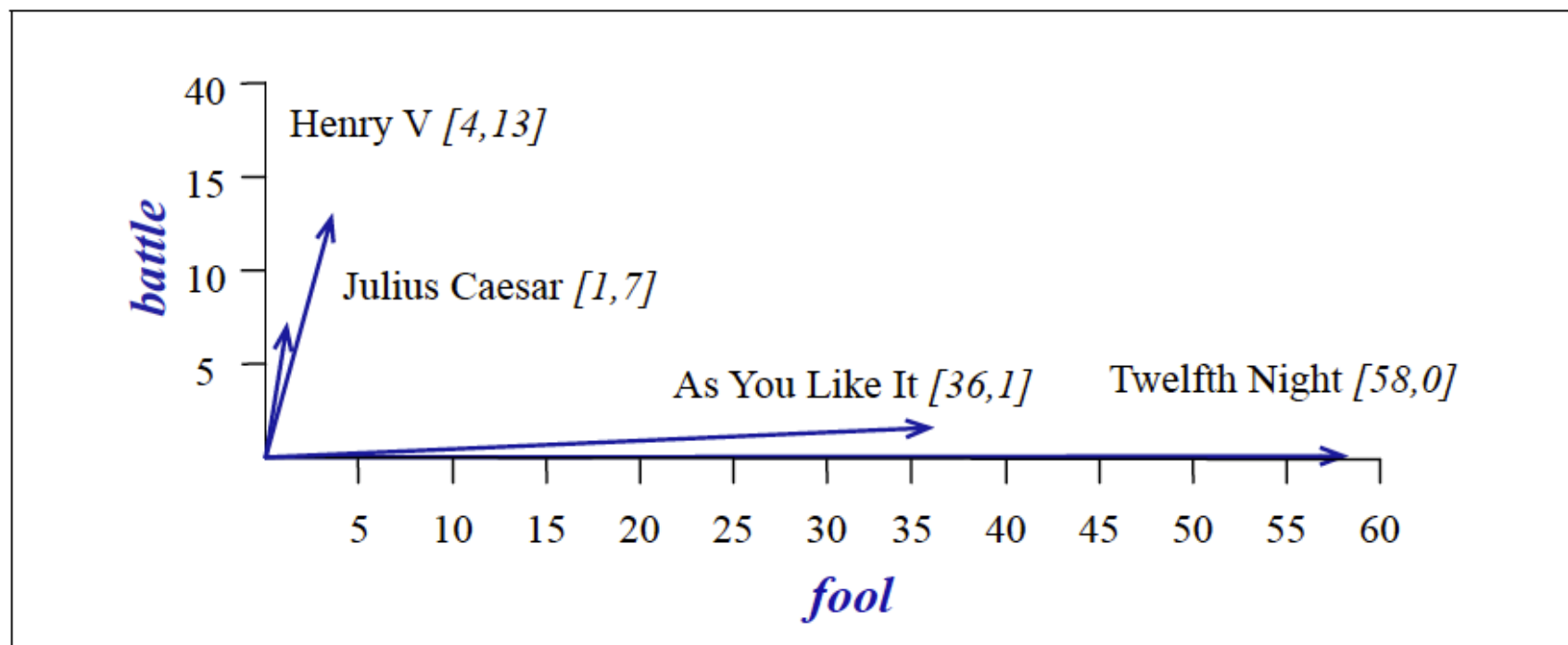
- Vector or distributional models of meaning are generally based on a co-occurrence matrix
 - ✓ term-document matrix
 - ✓ term-term matrix

Term-Document Matrix

	As You Like It	Twelfth Night	Julius Caesar	Henry V
battle	1	0	7	13
good	114	80	62	89
fool	36	58	1	4
wit	20	15	2	3

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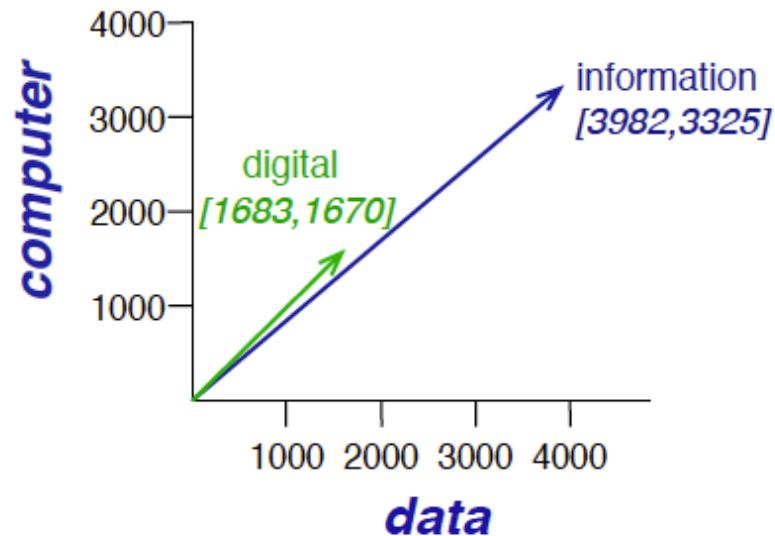
	As You Like It	Twelfth Night	Julius Caesar	Henry V
battle	1	0	7	13
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Term-Term Matrix

is traditionally followed by **cherry** pie, a traditional dessert
often mixed, such as **strawberry** rhubarb pie. Apple pie
computer peripherals and personal **digital** assistants. These devices usually
a computer. This includes **information** available on the internet

	aardvark	...	computer	data	result	pie	sugar	...
cherry	0	...	2	8	9	442	25	...
strawberry	0	...	0	0	1	60	19	...
digital	0	...	1670	1683	85	5	4	...
<u>information</u>	0	...	3325	3982	378	5	13	...

Term-Term Matrix



Term-Term Matrix

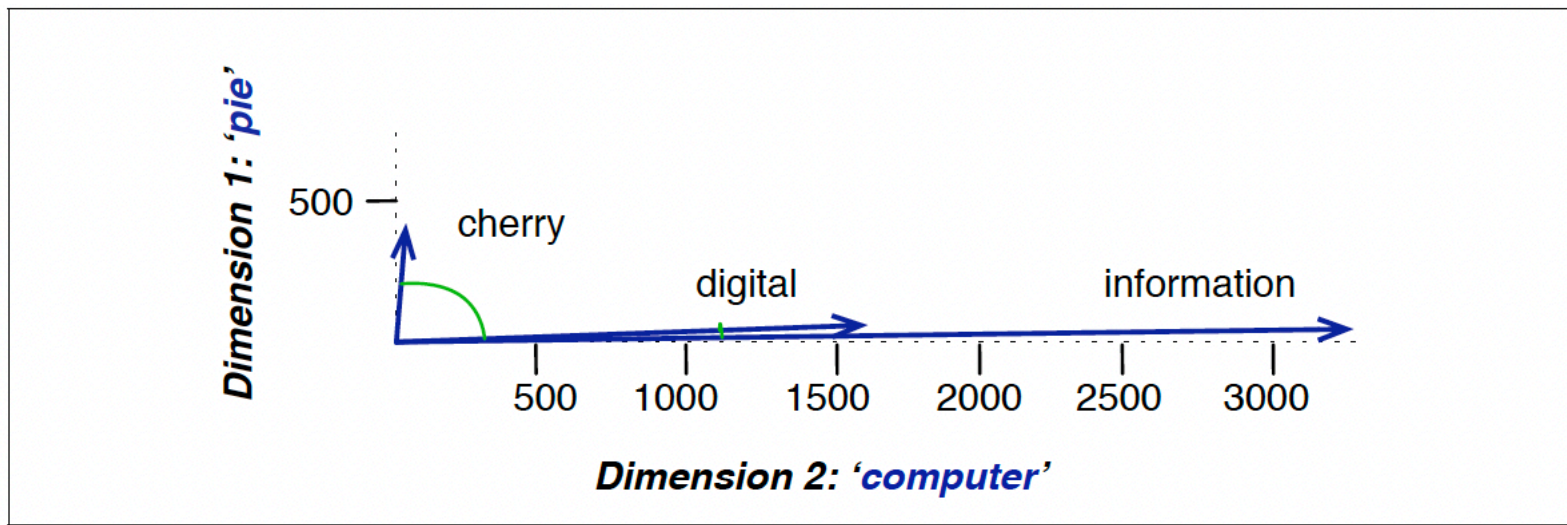
	pie	data	computer
cherry	442	8	2
digital	5	1683	1670
information	5	3982	3325

cos (cherry, information)= ??
cos (digital, information)= ??

$$\cos(\text{cherry}, \text{information}) = \frac{442 * 5 + 8 * 3982 + 2 * 3325}{\sqrt{442^2 + 8^2 + 2^2} \sqrt{5^2 + 3982^2 + 3325^2}} = .018$$

$$\cos(\text{digital}, \text{information}) = \frac{5 * 5 + 1683 * 3982 + 1670 * 3325}{\sqrt{5^2 + 1683^2 + 1670^2} \sqrt{5^2 + 3982^2 + 3325^2}} = .996$$

Term-Term Matrix



Pointwise Mutual Information (PMI)

- **Intuition**

- ✓ ask how much more the two words co-occur in the target corpus than we would have a priori expected them to appear by chance.
- The pointwise mutual information between a target word ***w*** and a context word ***c*** (Church and Hanks 1989-90)

$$\text{PMI}(w, c) = \log_2 \frac{P(w, c)}{P(w)P(c)}$$

- PMI values range from negative to positive infinity.

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$$\text{PMI}(w, c) = \log_2 \frac{P(w, c)}{P(w)P(c)}$$

- PMI values range from **negative** to positive infinity.
- negative PMI values(which imply things are co-occurring less often than we would expect by chance) tend to be unreliable unless our corpora are enormous

Positive Pointwise Mutual Information (PPMI)

$$\text{PPMI}(w, c) = \max\left(\log_2 \frac{P(w, c)}{P(w)P(c)}, 0\right)$$

Positive Pointwise Mutual Information (PPMI)

- ✓ Co-occurrence matrix F with W rows (words) and C columns (contexts)
- ✓ f_{ij} gives the number of times word w_i occurs with context c_j .
- ✓ Turn Co-occurrence matrix F into a PPMI matrix where
 - ✓ PPMI_{ij} gives the PPMI value of word w_i with context c_j

$$p_{ij} = \frac{f_{ij}}{\sum_{i=1}^W \sum_{j=1}^C f_{ij}}, \quad p_{i*} = \frac{\sum_{j=1}^C f_{ij}}{\sum_{i=1}^W \sum_{j=1}^C f_{ij}}, \quad p_{*j} = \frac{\sum_{i=1}^W f_{ij}}{\sum_{i=1}^W \sum_{j=1}^C f_{ij}}$$

$$\text{PPMI}_{ij} = \max\left(\log_2 \frac{p_{ij}}{p_{i*}p_{*j}}, 0\right)$$

Positive Pointwise Mutual Information (PPMI)

	computer	data	result	pie	sugar
cherry	2	8	9	442	25
strawberry	0	0	1	60	19
digital	1670	1683	85	5	4
information	3325	3982	378	5	13

$P(w=\text{information}, c=\text{data}) = ?$

$P(w=\text{information}) = ?$

$P(c=\text{data}) = ?$

$\text{PPMI}(\text{information}, \text{data}) = ?$

Positive Pointwise Mutual Information (PPMI)

	computer	data	result	pie	sugar	count(w)
cherry	2	8	9	442	25	486
strawberry	0	0	1	60	19	80
digital	1670	1683	85	5	4	3447
information	3325	3982	378	5	13	7703
count(context)	4997	5673	473	512	61	11716

$$P(w=\text{information}, c=\text{data}) = \frac{3982}{11716} = .3399$$

$$P(w=\text{information}) = \frac{7703}{11716} = .6575$$

$$P(c=\text{data}) = \frac{5673}{11716} = .4842$$

Positive Pointwise Mutual Information (PPMI)

	p(w,context)					p(w)
	computer	data	result	pie	sugar	p(w)
cherry	0.0002	0.0007	0.0008	0.0377	0.0021	0.0415
strawberry	0.0000	0.0000	0.0001	0.0051	0.0016	0.0068
digital	0.1425	0.1436	0.0073	0.0004	0.0003	0.2942
information	0.2838	0.3399	0.0323	0.0004	0.0011	0.6575
p(context)	0.4265	0.4842	0.0404	0.0437	0.0052	

$$\text{PPMI}(\text{information}, \text{data}) = \log_2(.3399 / (.6575 * .4842)) = .0944$$

	computer	data	result	pie	sugar
cherry	0	0	0	4.38	3.30
strawberry	0	0	0	4.10	5.51
digital	0.18	0.01	0	0	0
information	0.02	0.09	0.28	0	0

- ✓ *cherry* and *strawberry* are highly associated with both *pie* and *sugar*
- ✓ *data* is mildly associated with *information*

PPMI - Issues???

- ✓ Biased toward infrequent events
- ✓ Very rare words tend to have very high PPMI values

$$\text{PPMI}_\alpha(w, c) = \max(\log_2 \frac{P(w, c)}{P(w)P_\alpha(c)}, 0)$$

$$P_\alpha(c) = \frac{\text{count}(c)^\alpha}{\sum_c \text{count}(c)^\alpha}$$

Why should this work ??

Levy et al. (2015) found that a setting of $\alpha = 0.75$ improved the performance of embeddings on a wide range of tasks

PPMI - Issues???

- ✓ Biased toward infrequent events
- ✓ Very rare words tend to have very high PPMI values
- ✓ Reduce this bias toward low-frequency

$$\text{PPMI}_\alpha(w, c) = \max(\log_2 \frac{P(w, c)}{P(w)P_\alpha(c)}, 0)$$

$$P_\alpha(c) = \frac{\text{count}(c)^\alpha}{\sum_c \text{count}(c)^\alpha}$$

Why should this work ??

Hints: raising the count to $\alpha = 0.75$ increases the probability assigned to rare contexts, and hence lowers their PPMI ($P_\alpha(c) > P(c)$ when c is rare)

Word2vec

- ✓ vectors are **dense**
- ✓ the values will be **real-valued numbers** that can be negative
- ✓ proposed by [Mikolov et al. \(2013a, 2013b\)](#).
- ✓ skip-gram with negative sampling (SGNS)
- ✓ **static** embedding
- ✓ Code - <https://www.geeksforgeeks.org/python-word-embedding-using-word2vec/>